

Hong-Sheng Lai

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Education

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA

Master of Science in Computational Biology

Aug 2024 – Dec 2025

- Relevant Courses: Advanced Natural Language Processing, Generative AI, Generative AI in Biomedicine, Machine Learning, Applied Deep Learning, String Algorithm, Algorithm and Advance Data Structure

National Taiwan University

Taipei, Taiwan

Bachelor of Science in Biomechatronics Engineering, Minor in Economics

Sep 2019 – Jun 2023

- Relevant Courses: Data Structures and Algorithms, Introduction to Data Science, Data Analysis and Machine Learning with Python, Linear Algebra, Statistics, Database Management Systems, Computer Programming Language

Skills

- **Programming Languages:** Python, C++, Java, SQL, R, LaTeX
- **MLOps & Framework:** PyTorch, JAX, Grafana, Wandb
- **Developer Tools:** Git, Linux, Docker, Slurm, Kubernetes, CI/CD, MySQL, MongoDB, Apache Airflow

Experience

Genentech

South San Francisco, CA

Computational Biology Intern

Jan 2026 – Present

- Architected an automated RNA sequencing analysis pipeline for CRISPR-Cas9 gene editing and cell therapy.
- Developed a cross-platform GUI using **Ona AI agent** to streamline on-premises executions for end-users.
- Performed on **High-Performance Computing (HPC)** environments, controlled by **Slurm** workload manager.

Computational Biology Department, School of Computer Science, CMU

Pittsburgh, PA

Graduate Researcher

Jan 2025 – Dec 2025

- Engineered a controllable codon design model by applying **activation engineering** techniques to both **decoder-only** and **encoder-decoder language models** to achieve multiple objectives optimization.
- Architected a multi-objective **reinforcement learning** framework for mRNA sequence optimization, implementing **demonstration-guided** replay and **milestone-based reward** shaping.

Taiwan AI Labs

Taipei, Taiwan

Machine Learning Engineer Intern

Jul 2023 – Jan 2024

- Enabled variant similarity search in **Qdrant vector database** with **contrastive learning and genomic language models**.
- Accelerated sparse matrix regression models with **JAX**, significantly improve computational speed for **federated analysis**.
- Established an end-to-end clinical annotation pipeline for pharmacogenomics for drug usage recommendation.
- Automated model deployment, testing, and monitoring pipelines using **GitLab CI/CD, Kubernetes, and Grafana**.

Bioinformatics and Machine Learning Lab, NTU

Taipei, Taiwan

Undergraduate Researcher

Jun 2021 – Jul 2024

- Evaluated the structural impact of genetic variants with **protein structure prediction** tools (e.g., AlphaFold2, ESMFold)
- Designed an alignment-based algorithm to accurately estimate copy number variations and call variants in complex gene regions (e.g., pharmacogenes, KIR, HLA) from haplotype assemblies and long-read sequencing data.

Missemi Intelligence CO., LTD.

Taipei, Taiwan

Software Engineer, Part-time

Jun 2021 – Jun 2023

- Built and launched a short-read/target sequencing next-generation sequencing (NGS) gene reporting product, driving commercial success with over 400 units sold.
- Automated PDF reports by integrating Python and HTML templates, using workflow management **Apache Airflow**, **Docker** containerization, and **command-line interface (CLI)**.

Projects

Enhancing Long-Context Dependent LLMs for Structure-Aware Data

Aug 2025 – Dec 2025

- Evaluated the architectural limitations of **long-context** LLMs in effectively capturing non-sequential document structures.
- Developed an **RAG** system utilizing **hybrid retrieval, reranking**, and advanced **prompt engineering** to overcome long-distance context limitations.

Do We Need the Whole Genomic Sequence in Genomic Language Models?

Jan 2025 – May 2025

- Applied **adversarial testing** techniques to assess the robustness of genomic language models.
- Developed a compression method achieving 18% higher compression than random masking with equivalent model performance.